# Manual Calculation.

Exp 1:

1. Program A:

8-bit addition & subtraction.

A = 5AH , B = 27H

Addition:

5AH + 27H = 81H

Sum = 81H, CY = 0
AC = 1 , P = 1 , Z = 0 , S = 1

Flag byte = 94H
Signed overflow = Yes

Subtraction:

5AH − 27H = 33H

Diff = 33H
Borrow = 0, AC = 0
P = 1, Z = 0, S = 0

Flag byte = 06H
Signed overflow = NO

2 Program B:

16-bit Addition & Subtraction.

X = 1234H , Y = 4321H

Add:

1234H + 4321 = 5551H
Carry 0

Sub:

1234 − 4321 = CF13H
Borrow = 1

3. Program C :

8-bit multiply.

14H × 06H

Division F2H ÷ 0AH

Multiply : 14H × 06H

= 0678H

Division :

F2H ÷ 0AH

Quo = 18H

Rem = 02H

4. Program D :

16-bit Multiplication & Division.

Multiply : 1357H × 0025H

Division : 9A23H ÷ 0011H

Multiply : 1357H × 0025H

= 002CA13H

Upper word = 0002H ; Lower word = CA13H.

Division : 9A23H ÷ 0011H

Quo = 0911H

Rem = 0002H